

KAILAS KUMAR A

AI/ML Engineer

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Kochi, Kerala, India

PROFILE

Results-oriented AI/ML Engineer with a strong foundation in predictive modeling and computer vision. Proven ability to transform raw data into actionable insights through rigorous EDA and feature engineering. Dedicated to building efficient, scalable machine learning solutions that solve real-world problems and optimize decision-making processes.

SKILLS

Programming Languages:	Python, C++, SQL
Deep Learning Frameworks:	TensorFlow, Keras, MediaPipe
Libraries:	NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, Seaborn
Generative AI:	LLM Implementation (Gemma) (Basic), RAG Pipeline Development, Prompt Engineering (Basic)
Cloud (AWS):	EC2, S3, MFA / IAM Security
Tools:	Jupyter, VS Code, Git
Soft Skills:	Analytical Thinking, Technical Communication, Problem Solving, Teamwork

EXPERIENCE

Data science Intern <i>Luminar Technolab</i>	July 2025 – March 2026 <i>Kochi, Kerala</i>
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- Developed and applied Machine Learning models using Scikit-learn and Pandas for predictive data analysis
- Performed Exploratory Data Analysis (EDA) and data visualization using Matplotlib and Seaborn to identify key trends
- Cleaned and preprocessed large datasets to ensure high data quality for model training and validation
- Presented technical findings and data insights during weekly research meetings to improve project outcomes

Python internship program <i>Softloom IT Solutions</i>	April 2024 <i>Kochi, Kerala</i>
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- Mastered core Python development workflows and technical operations through an intensive software internship.
- Executed deep-dive research into division-specific modules to implement complex subject matter and technical solutions.
- Demonstrated proactive problem-solving and high-quality technical output to meet diverse team and project objectives.

PROJECTS

Student Dropout Prediction System <i>Python, NumPy, Pandas, Matplotlib, Seaborn</i>	Machine Learning Project
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- Project Link
- Built a classification model using the OULAD dataset to identify at-risk students among 32k+ records
 - Engineered behavioral features from VLE clickstream data to predict withdrawals without using grade history
 - Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) using Python, Pandas, and NumPy
 - Executed Random Forest model using Scikit-learn to improve prediction performance
 - Optimized model performance to achieve an 80% accuracy score in identifying at-risk students

Air Canvas Gesture Drawing Application <i>Python, OpenCV, MediaPipe, NumPy</i>	Computer Vision Project
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- Project Link
- Developed a real-time computer vision application allowing users to draw on a virtual canvas using hand gestures captured via webcam
 - Engineered robust gesture recognition utilizing Google's MediaPipe Hands framework to track 21 3D hand landmarks
 - Calculated Euclidean distances between thumb and index landmarks to dynamically map and scale brush sizes in real-time
 - Performed matrix operations and image thresholding using OpenCV and NumPy to blend the digital canvas overlay
 - Designed application state-management features, including a multi-step undo history stack and local image saving

EDUCATION

Bachelor of Vocation (B.Voc) in Software Development <i>University of Kerala, Thiruvananthapuram</i>	2022 – 2025 <i>CGPA 6.5</i>
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CERTIFICATIONS

Certification in Data Science-Python <i>Luminar TechnoLab, Ernakulam</i>	July 2025
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